



Republic of the Philippines
**Polytechnic University of the
Philippines**
College of Engineering
Computer Engineering Department

Embedded Systems

Smart Cat Feeding and Monitoring Station

Introduction

a. The Team

BSCOE 4-2 / Group 5

Name	Role
Bugnon, Jelisha Ruth B.	Prototype Designer
Cando, Irish Micole T.	Website Developer
Corpuz, Angela E.	Prototype Designer
Dandamun, Mahid L.	Hardware Engineer
Neri, Miguel Alfonso B.	Prototype Designer

b. Overview of the system

In modern urban lifestyles, many cat owners lead busy schedules that require them to be away from home for extended periods, whether due to work commitments, frequent travel, or other obligations. While cats are often considered low-maintenance pets compared to dogs, they still require regular care, particularly in terms of feeding and health monitoring. The absence of a reliable and consistent feeding routine can negatively affect a cat's well-being, especially when owners are unable to be present at scheduled mealtimes.

Irregular feeding patterns may lead to a variety of health issues in cats, such as obesity, malnutrition, or gastrointestinal discomfort. These issues are often exacerbated when there is no one available to monitor the pet's behavior or appetite changes, which can be early signs of underlying medical conditions (Gunn-Moore, 2005). Moreover, prolonged solitude without supervision can result in stress and behavioral problems for the cat. For owners, this lack of oversight often results in anxiety and uncertainty about their pet's condition while they are away.

The Smart Cat Feeding and Monitoring Station is an embedded system designed to provide convenience and assurance for cat owners who are frequently away from home. It automates feeding schedules, monitors food levels, and detects cat presence and feeding behavior. With a combination of sensors, actuators, and a microcontroller, the system ensures timely feeding while tracking essential metrics through a web-based dashboard. This setup not only promotes better cat care but also allows owners to monitor and respond remotely.

c. Objectives and goals of the system

The main objectives of the Smart Cat Feeding and Monitoring Station are as follows:

- To **automate cat feeding** on a schedule using a servo-controlled dispenser.
- To **monitor food intake and cat presence** using weight sensors and motion detectors.
- To **alert the owner** when the cat is not eating, food levels are low, or other abnormal activities are detected.
- To **log and display feeding data** and cat presence via an intuitive web dashboard.
- To provide **manual feed control**, both physically through push buttons and remotely through the dashboard.

d. Features of the system

The following are the features implemented into the system, ensuring full delivery of the objectives mentioned beforehand:

- **Feeding Detection:** Measures weight before and after feeding to log intake.
- **Cat Presence Monitoring:** Uses PIR sensor to track cat approach and interaction time.
- **Real-Time Alerts:** Buzzer, RGB LED, and web notifications for low food or missed feeding.
- **Web Dashboard Integration:** Displays feeding logs, activity data, and manual control options.
- **LCD Display:** Provides on-device information including food weight and system alerts.
- **Scheduled and Manual Feeding:** Automated feeding based on real-time clock, with options for physical or remote manual feed override.

Hardware Requirements

a. List of necessary components needed/used. Discuss briefly each component and its purpose.

Quantity	Component Name	Purpose
1	ESP32 WROOM 32D (38pin)	Main microcontroller with Wi-Fi/Bluetooth for connectivity and logic control
1	Ultrasonic Sensor (HC-SR04)	Detects the distance of food level
1	PCB for power and modules	For power distribution and connection of components
1	PIR SENSOR	Detects the presence of animal via heat
1	Load Cell + HX711 Amplifier	Measures the weight of the food bowl to detect feeding
1	Servo Motor	Dispenses food automatically

1	Real-Time Clock (RTC) Module	Keeps accurate time for feeding history
1	Buzzer	Gives sound alerts for low food
1	RGB LED	Visual indicator for low food
1	LCD	Displays real-time information such as current food weight/percentage, feeding status, alerts
2	Momentary Push Button	Buttons for dispensing and refilling of the food tank
1	Food Bowl	Container where the cat eats; used in conjunction with the load cell to monitor food intake
1	Food Container	Storage container for cat food; supplies food to the dispensing mechanism
—	Plywood	Used as the primary material for building the casing and structural frame

Software Requirements

a. Libraries required for the project (if any)

- marcoschwartz/LiquidCrystal_I2C@^1.1.4
- bogde/HX711@^0.7.5
- knolleary/PubSubClient@^2.8
- bblanchon/ArduinoJson@^6.21.3
- madhephaestus/ESP32Servo@^0.13.0
- makuna/RTC@^2.5.0

b. Source Code (C++ program). Please provide a google link of the sketch

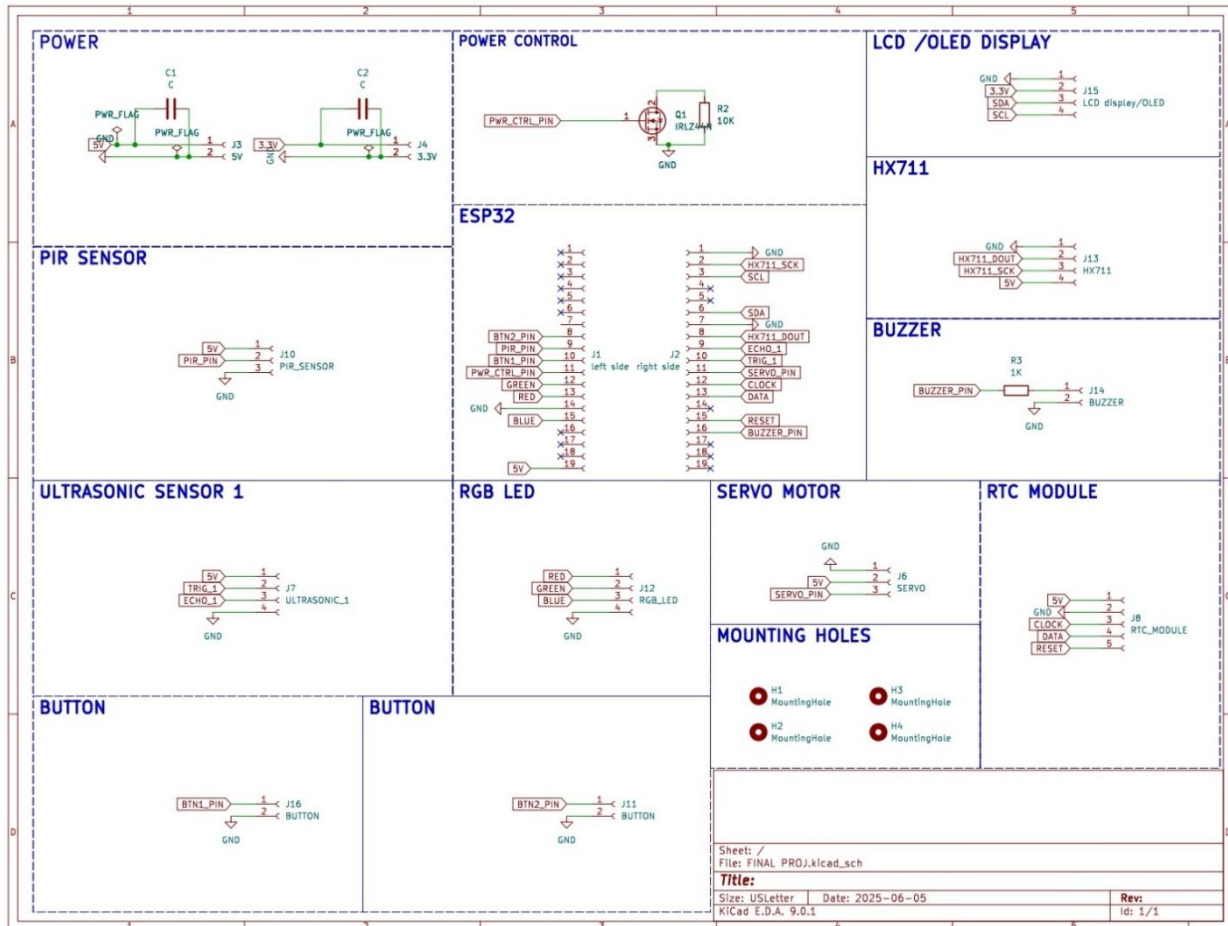
GitHub Repository Links for the source codes used in developing the system:

1. **Hardware Code:** <https://github.com/MahidDandamun/Embedded-System-Final-Project.git>
2. **Website Code (Front End):** <https://github.com/irishmicoletcando/pet-feeder-frontend.git>
3. **Website Code (Back End):** <https://github.com/irishmicoletcando/pet-feeder-backend.git>
4. **Actual Website:** <https://pet-feeder-ruddy.vercel.app>

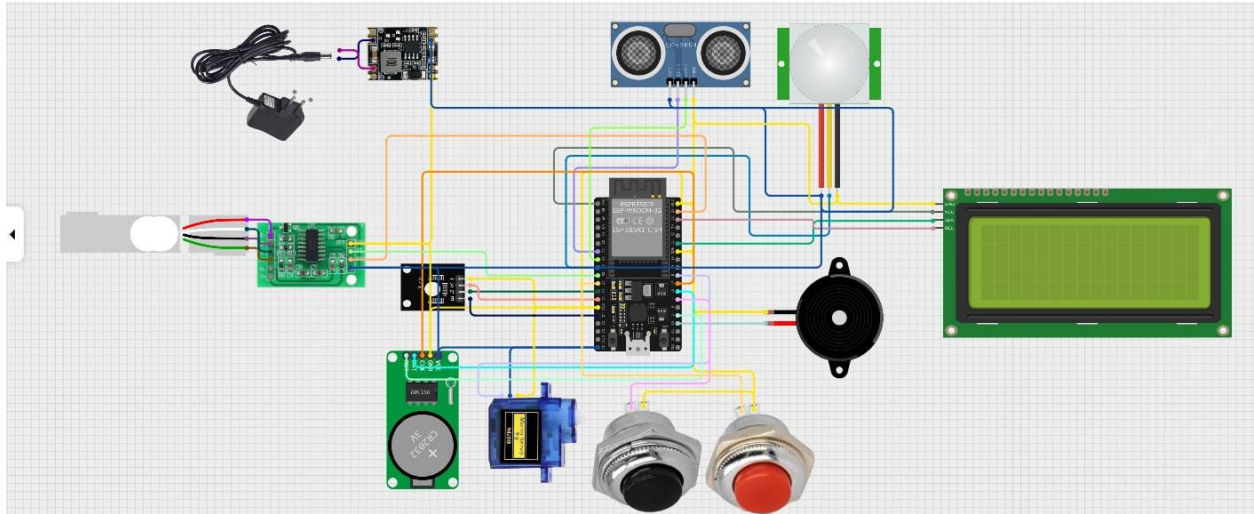
System Design and settings

a. System Architecture/Interfacing diagram of the system

Schematic Diagram



Interface Diagram

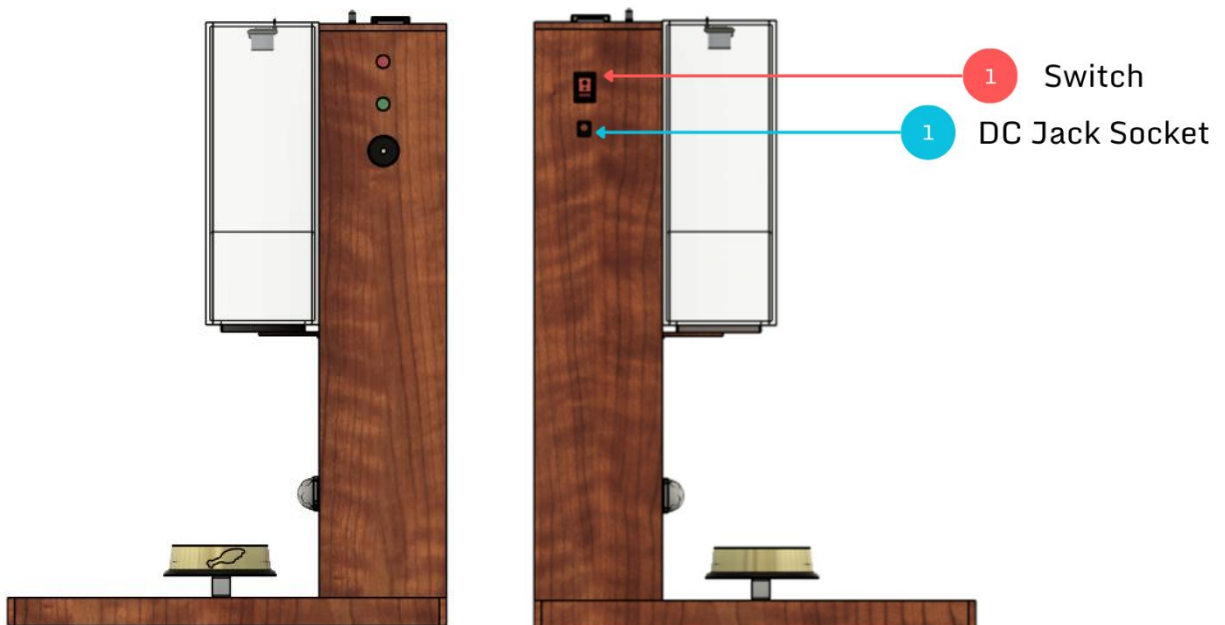
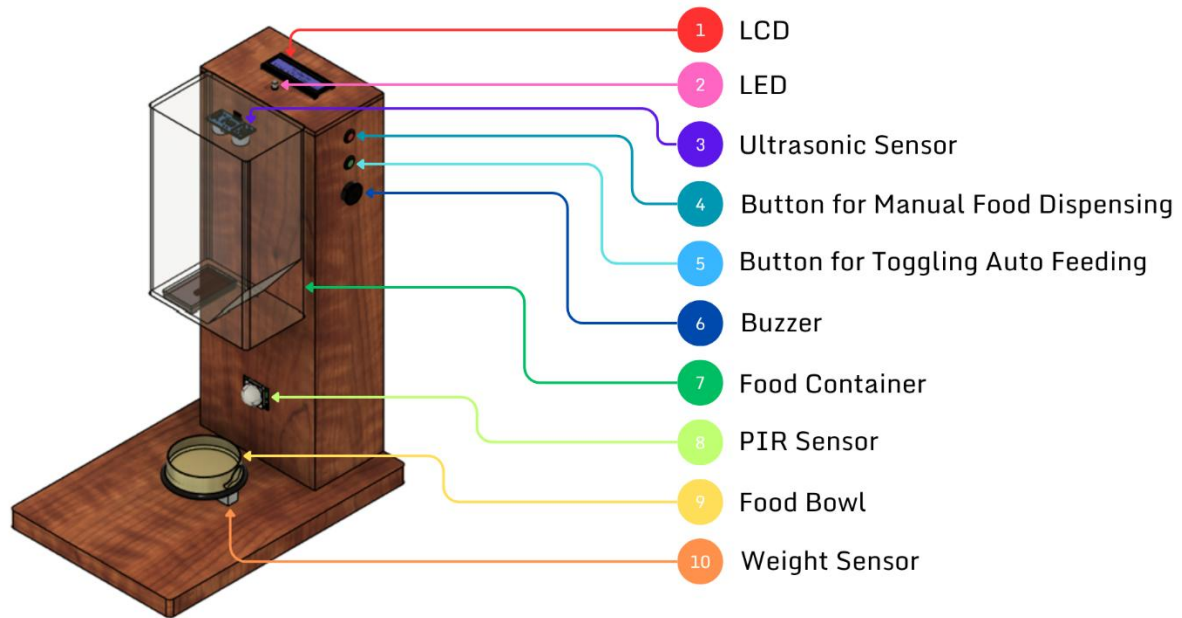


Current Consumption

Component	Current (mA)
Ultrasonic Sensor	6
Servo Motor	500
ESP32 (WiFi)	400
RGB Module	20
LCD I2C	3
PIR Sensor	3
RTC Module	0.3
HX711	1.6
Buzzer	30
Buttons (est.)	1
Total	964.9

b. The Prototype Design

3D Model of the Prototype





Innovation

a. Innovation of the projects

- **Cat Presence Detection**

By integrating a Passive Infrared (PIR) sensor, the system can detect when the cat is near the feeding station. This enables tracking of cat behavior and feeding patterns.

- **Website Integration**

A custom web dashboard allows cat owners to remotely monitor feeding activity, food levels, and cat presence. It also provides real-time alerts and historical data, offering convenience and peace of mind, especially for owners who are frequently away.

- **Manual Override through Button and Web**

To increase flexibility, the system supports manual food dispensing both physically via push buttons and remotely through the web interface. This dual-mode override ensures owners can respond to unexpected situations anytime, from anywhere.

Conclusion

a. Summary of the system

The Smart Cat Feeding and Monitoring System aims to provide convenience for cat owners who are frequently away from home through automated feeding schedules, monitoring cat's food levels, and detecting cat presence and feeding behavior. Its main features consist of logging feeding intake into the database, real life monitoring of cat presence through PIR sensor, real time alerts for low food level or missed feeding schedule, web dashboard for overview of activity data along with manual control options, an LCD display for providing device information such as food weight and system alerts, and automated or manual feeding based on real time clock.

The Smart Cat Feeding and Monitoring System mainly consists of ESP32 microcontroller for controlling the entire system, PIR sensor for cat presence detection, Ultrasonic Sensor for determining the food level, Load Cell for measuring the weight of food on the bowl, Servo Motor for dispensing the food automatically, and the combination of Buzzer, RGB LED, and LCD for displaying information and sending alert when the food level is low.

To further enhance the system, a PIR sensor was integrated so that the system can detect when the cat is near the system, allowing tracking of cat behavior and its feeding patterns. Website integration was also utilized to further improve as it allows users to remotely monitor any activity and food levels through the dashboard, offering convenience and peace of mind for cat owners. Furthermore, a manual override was also implemented through a physical button and in the web application, allowing manual food dispensing so that users can still respond to unexpected situations anywhere and anytime.

Overall, the system was able to perform its intended functions effectively, dispensing food accurately during user-defined feeding times set through the web application. The integration of the PIR sensor allowed reliable detection of cat presence near the feeding station, enabling monitoring of feeding behavior and interaction patterns. Real-time alerts through the buzzer, RGB LED, and web notifications ensured the user was promptly informed of low food levels or missed feedings, adding an extra layer of reliability. The web dashboard provided a clear and intuitive interface for monitoring system status and feeding logs, while also supporting remote manual control, offering convenience and flexibility. The LCD display on the device ensured that essential information like food weight and system alerts were always visible on-site. Together, these components created a cohesive and responsive system capable of automating cat feeding while maintaining real-time awareness of both food supply and cat activity.

b. Knowledge gained / Lessons learned

Through the Smart Cat Feeding and Monitoring System, we have gained valuable experience in hardware and software integration, resulting in the creation of a cohesive and user-centric IoT solution. We realized the importance of designing systems that prioritize user accessibility, provide clear and timely system feedback, and maintain a high level of reliability, especially in real-world applications where consistency and responsiveness are critical.

The project also taught us how crucial it is to plan for both expected and unexpected user interactions, which led to implementing manual override options both physically and remotely. We learned the significance of real-time data processing and communication between components, especially in synchronizing sensor inputs, actuator responses, and web-based updates. Additionally, working with multiple sensors and modules required us to address challenges in power management, data accuracy, and component compatibility, further strengthening our troubleshooting and system optimization skills.

Furthermore, this project helped us understand how to structure and present data meaningfully through a web dashboard, highlighting the role of user interface design in IoT systems. It also emphasized the need for modular and scalable code, enabling easier maintenance and future enhancements. Overall, this experience not only deepened our technical understanding but also reinforced the value of thoughtful design, thorough testing, and user-focused development in solving everyday challenges faced by cat owners.

c. Future enhancements and areas for improvement

While the Smart Cat Feeding and Monitoring System was able to do its functions along with the enhancements implemented, here are key ideas that will help to maximize the performance of the system:

1. **Live Camera Monitoring:** Integrate a camera to provide real-time video streaming through the web dashboard, allowing users to visually check on their cat's behavior and feeding activity remotely.
2. **Mobile App Integration:** Develop a dedicated mobile application for easier access to system controls, alerts, and real-time monitoring, offering greater convenience compared to a web-only interface.
3. **Multi-Pet Support:** Use RFID tags or image recognition to identify individual cats, enabling personalized feeding schedules and accurate tracking of food intake per cat.
4. **Weather-Resistant Design:** Upgrade the casing and structural materials to make the system durable for outdoor use, protecting it from rain, dust, and varying temperatures.
5. **AI-Based Feeding Analysis:** Introduce machine learning to analyze feeding patterns over time, helping detect irregular behavior or early signs of health issues in cats.

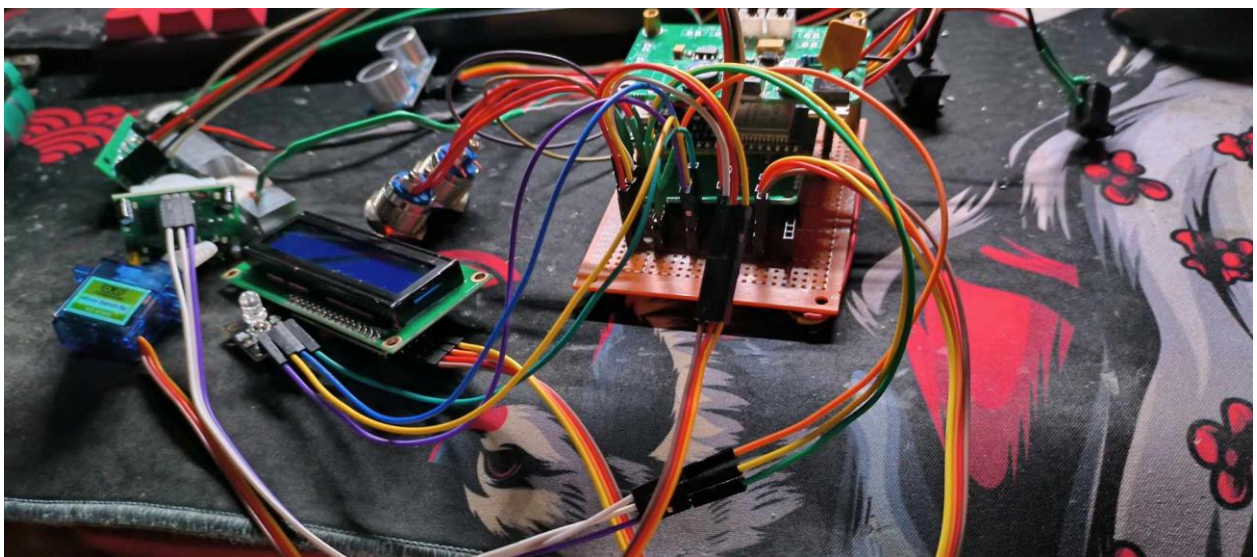
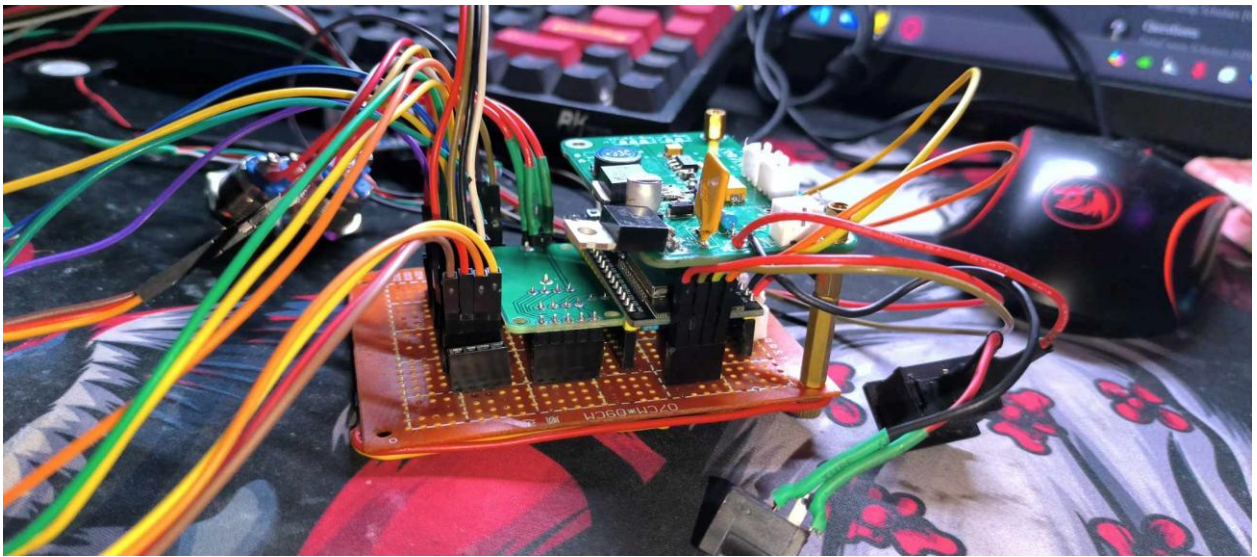
References and Documentation

a. List of references

Gunn-Moore, D. A. (2005). Caring for elderly cats: Recognizing and treating the signs of aging. *Journal of Feline Medicine and Surgery*, 7(5), 261–268. <https://doi.org/10.1016/j.jfms.2005.06.001>

b. Photo documentation

Assembling and Troubleshooting of Hardware Components



Actual Prototype (Front View)



Actual Prototype (Left and Right Side View)



Actual Prototype (Top View)



Actual Prototype (Back View)



Close Up View of the Components



Sign In Page of the Website



PetFeeder Pro

Keep your furry friends happy and healthy with smart feeding monitoring

🐾 Smart • Safe • Simple 🐾

[Sign In](#)[Sign Up](#)

Welcome Back!

Sign in to monitor your pet's feeding

Email

Password

[Sign In](#)

Sign-Up Page of the Website



PetFeeder Pro

Keep your furry friends happy and healthy with smart feeding monitoring

🐾 Smart • Safe • Simple 🐾

[Sign In](#)[Sign Up](#)

Join PetFeeder Pro!

Create an account for you and your pet

Your Name

Pet Name

Email

Password

[Create Account](#)

Dashboard Page of the Website

Welcome back! How's Bochoh doing? 🐾

Monitor your pet's feeding and activity in real-time

Last updated: 11:00:00 AM ↺

Daily Total Food

195g

Food Container

Full

100% Capacity

Pet Status

Active

Pet detected nearby

Manual Feed

Feed Now

Ready to Dispense

Container Level: 100%

Feed Now



Profile Page of the Website

 **PetFeeder Pro**

[Dashboard](#) [Profile](#)



Irish Micole Cando
Pet Parent

 irishmicolecando@gmail.com

 Member since June 2025

 **Pet Information & Stats**
Your furry friend's details and feeding statistics

[Refresh](#)



Bochog
Happy & Healthy

42
Total Feedings

5.3g
Daily Average

1209.5g
Food Consumed

8
Days Active

38
of Feedings This Week

 **Account Settings**

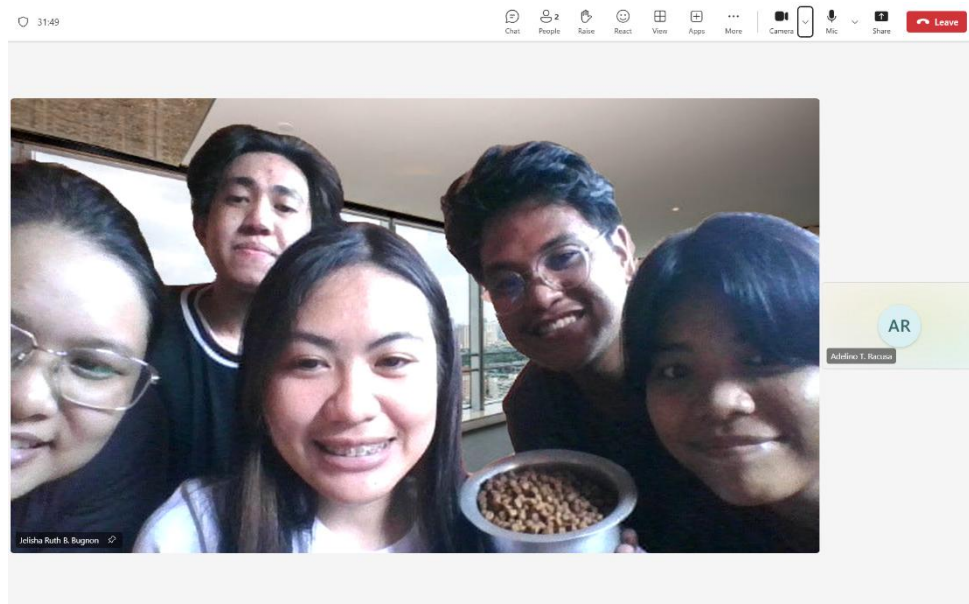
Sign Out
Sign out of your PetFeeder Pro account

[Logout](#)

Finalization of the Overall System



Group Photo After Presentation

**Final Grade**